

AIDAN M. MOKALLA

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EDUCATION GPA: 3.7/4

Reed College, Portland, Oregon, USA, 2025

- + Spanish minor, sole member of graduating class with double majors: Computer Science & Linguistics,
- + A grades in *Real Analysis*, *Linear Algebra*, *Deep Learning*, *Linguistic Field Methods*, and *Psycholinguistics*,

Bryn Mawr College, Dushanbe, Tajikistan, 2024

- + Fellowship funded by international non-profit *American Councils* to research Persian language varieties,

La Universidad de Carlos III and **Middlebury College**, Madrid, Spain, 2023

- + Spanish-language courses including *Artificial Intelligence*, *Data Science*, and *International IP Law*.

PUBLICATION

Source of the Storm: Interpreting Weather Models to Detect Cyclogenesis, *forthcoming*

- + Applying post-superposition statistical interpretation methods to weather prediction models to improve localization of rare climatic events and demonstrate methodological generalizability beyond transformer-based models,

Evaluating Diffusion Language Models Beyond Benchmarks COLM 2026

- + Found that people, when evaluating the outputs of the recently invented *diffusion-based* LLM architecture, consistently over-trust AI outputs at a significantly lower rate than when evaluating outputs of traditional LLMs,

Interpretable Signals Reveal Failure-Relevant Representations in VLA Models, ICML 2026

- + Discovered representational mechanistic predictors of vision-language-action model task failure,

Fault- and Network-Agnostic Agreement, *with* Erica Blum, TCC 2026

- + Proved tight running-time bounds for all algorithms achieving network-agnostic Byzantine consensus,

Language-Agnostic Concept Spaces Found in Large Language Models, *with* Greg Anderson, *thesis*, 2025

- + Designed, implemented, and evaluated novel architecture to interpret transformer-based LLMs by leveraging natural-language translation of ground-truth semantics. My approach learns convergent “concept spaces” over LLM activations in parallel, yielding more consistent LLM interpretations than prior work,

Modulated Signal Filtering: Wouldn't it be Noise?, *with* Sameer ud Dowla Khan, *thesis*, 2025

- + Collected over 3,000 participant responses to investigate how listeners use unique cognitive strategies to parse spoken speech versus to parse song lyrics, using both deep and symbolic statistical methods,
- + Demonstrated that we use contextually-differentiated mechanisms to filter noise from the “linguistic signal” by making significantly more accurate predictions of how people interpret unclear speech under this prior.

LEADERSHIP & EXPERIENCE

Algoverse AI Research, *Machine Learning Researcher, Fellow, & Mentor* *September 2025 – present*

- + Investigating and publishing methods of explainable AI and climate science,

Linguistic Diversity Initiative, *Lead Volunteer Ambassador*, *June 2023 – August 2025*

- + Leading a team of three translators, I compiled and published dictionaries of terminology used in Portland Public’s science curricula, translated into 6 languages, for households with low English proficiency, already enabling over 800 families to engage in their students’ education,
- + Taught seminars to students and faculty about sociolinguistic research on language use in undergraduate settings,
- + Organized and lead twice-monthly meetings of over 50 volunteers and larger events,

Reed College Student Body, *Senator & Honor Councillor*, *September 2021 – May 2025*

- + Peer-elected to serve in faculty committees including the *Committee on Academic Policies and Planning* and *Committee on Computer Policy and Planning*, disburse 6-figure Student Body Fund and other resources to college organizations, and interview and hire all student applicants for on-campus roles,
- + Mediated and resolved conflicts between students, professors, and other members of the college community,

Stoel Rives, LLC, *Intern Practise Assistant*, *May – August 2022*

- + Implemented compliant automation of docketing, court appearances, and other events from firm-wide databases,
- + Maintained communication on behalf of attorneys while drafting and proofreading attorney correspondences.

SKILLS & ACCOLADES

Computational: Python, various data and NLP pipelines, PyTorch, NumPy, JS, Next.js, C++, PHP, SQL, Swift,

Awards: Presidential Commendation for Academic Excellence, Reed College Music Scholarship, National Diversity in STEM Grant, Department of State Critical Languages Scholarship,

Certifications: Federally eligible for non-competitive civil service, licensed in Conflict Mediation & Resolution, CPR,

Languages: Fluent Spanish, English, intermediate Farsi, Tajik, French, basic Portuguese, Dari, Chinese (Mandarin).